

products such as GE Fanuc's PACSystems are emerging as a key contender for both PLCs and PCs.

James Truchard, president and CEO of National Instruments, explains in a recent report that the future of PACs hinges on the incorporation of embedded technology. One example is the ability to use software to define hardware. Consider FPGAs. "FPGAs are electronic components commonly used by electronics manufacturers to create custom chips, allowing intelligence to be placed in new devices. These devices consist of three main components: configurable logic blocks that can perform a variety of functions, programmable interconnects that act as switches to connect the function blocks together and input-output (I/O) blocks that pass data in and out of the chip. By defining the functionality of the configurable logic blocks and the way they connect to each other and to the I/O, electronics designers can create custom chips without the expense of producing a custom ASIC (application-specific integrated circuit). FPGAs are comparable to having a computer that literally rewires its internal circuitry to run your specific application."

2 Mobile-enabled manufacturing is finally materialising

When everything is going mobile, why not manufacturing too? While there has been a lot of talk of 'mobile manufacturing' or mobile-enabled manufacturing in the past few years, it was in 2009 that the concept actually picked up steam. Mobile manufacturing is basically a situation where your industrial automation systems can be accessed and controlled via a mobile network, using a PDA, laptop or mobile phone. Applications, services, infrastructure, standards and most importantly proof-of-concepts, are all emerging, making it a real concept and not a distant dream.

As mentioned earlier, several manufacturers have also realised the importance of the grey-collared mobile work-

If this is v2, what will v3 be like?

Jitendra Kataria of Beckhoff says v3 will include use of:

1. Object Oriented Programming extension of IEC61131-3
2. Higher-level programming languages for programming PLC/motion control logic
3. Multi-core PC-based platforms which can do multiple tasks, viz, robotics, motion control and PLC, at the same time using different cores of the same PC-based system

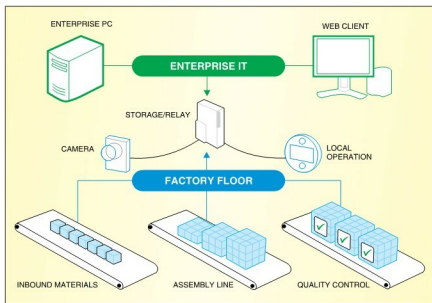


Fig. 4: The factory is at last becoming a part of the larger enterprise IT infrastructure

force, and invested in enhancing their role in the organisation. Some have adopted manufacturer-friendly applications from companies like SAP, Sybase, Microsoft, Capgemini, Wonderware and SAT Corporation, while others have developed their own mobile-enabled applications or added mobile capability to their existing systems. Having realised the power of the grey collar, manufacturers are even eagerly waiting for the roll-out of fourth-generation (4G) networks to invest further power in their hands—anytime, anywhere video conferencing, live demos and factory checks, remote troubleshooting by technicians and more.

Rosy as the picture seems, in truth the large number of mobile devices and applications are not easy to manage. Each device is a gateway to the organisation and security can be easily compromised if a device is lost. Solution providers are coming up with enhanced security and management

features to help manufacturers overcome this hurdle. One significant move in this direction is the subscription-based managed mobility service being offered by the Verizon-Sybase team. It provides an integrated framework of management and security services delivered via a Web portal. More such services are sure to come as more manufacturers go mobile.

3 Factories are becoming networked, and that too wirelessly

Without a doubt, networking has been one of the best things to happen to industrial automation in the past decade. Without Ethernet and IP, interoperability would have been still a distant dream. Networking brought about what can be called the design-to-product continuum. It has enabled a lot of aspects of industrial automation ranging from integration to diagnostics.

Now networking has become even